J1
USB4125-GF-A
USB_POWER
U6
TPS259474LRPWR
U3
BQ25173DSGR
U5
TPS61023DRLR
U19
TPS2116DRLR
U4
TPS62132RGTR
U7
TPS7A0225PDBVR
U40
TPS2116DRLR
Actual qualified control
File: control.kicad_sch
actual150ohm raw-source feed
#FLG5

OWNER ACCEPTED:5V USB-C advertising1.5A or3A. UFP only; noUSBdata/noPD escalation. GCT six contacts, not old24pin map.

RawCC/independentAON never fed by backup. TPS259474L is a circuit-breaker/latch-off; no guaranteed transient current ceiling.

SOURCE_SENSE_OD is physical wired-AND rawUV +USBUV +eFusePG. Status adds actualbuckPG and12..28ms assertion delay.

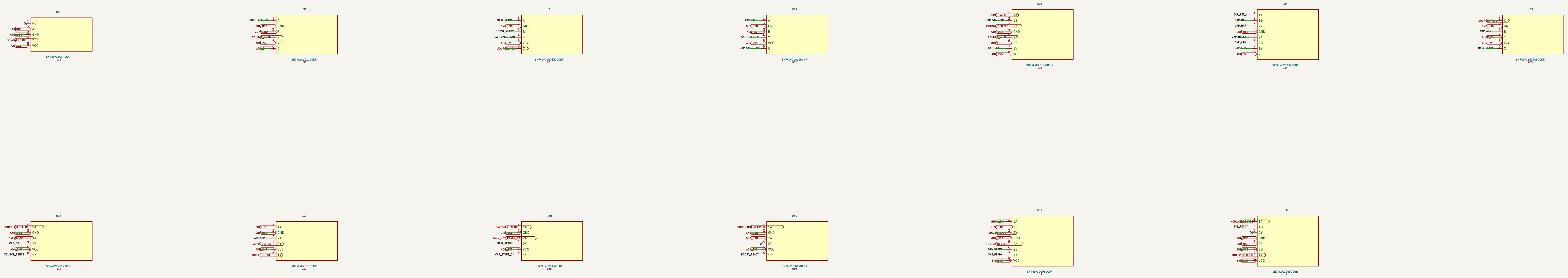
One cap-arm NAND latch: loadedUV or loss of both source and ready backup clears; rebound alone cannot rearm.

SeparateBQ nominal25mA charger. Cap-onlyTPS61023; measured current/ESR/thermal/USB/mechanical/routing gates remain open.

Sheet: /
File: qualified15.kicad_sch

Title: NOVA qualified >=1.5A prototype – main

Size: A1 Date:
KiCad E.D.A. 10.0.7
Rev: Q15-PROTOTYPE-1
Id: 1/2


U28
SN74LVC1G14DCKR
U30
SN74LVC1G11DCKR
U31
SN74LVC1G0832DCKR
U32
SN74LVC1G11DCKR
U33
SN74LVC2G132DCUR
U34
SN74LVC2G132DCUR
U35
SN74LVC1G3208DCKR
U36
SN74LVC2G17DCKR
U37
SN74LVC2G17DCKR
U38
SN74LVC2G14DCKR
U39
SN74LVC2G14DCKR
U17
SN74LVC2G08DCUR
U18
SN74LVC2G08DCUR

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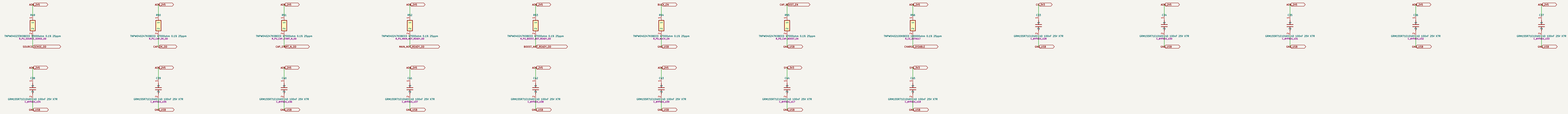
R49
TNPW040233K0BEED 33000ohm 0.1% 25ppm
R_PU_SOURCE_SENSE_OD
R50
TNPW040247K0BEED 47000ohm 0.1% 25ppm
R_PU_CAP_OK_OD
R51
TNPW040247K0BEED 47000ohm 0.1% 25ppm
R_PU_CAP_START_N_OD
R52
TNPW040247K0BEED 47000ohm 0.1% 25ppm
R_PU_MAIN_NOT_READY_OD
R53
TNPW040247K0BEED 47000ohm 0.1% 25ppm
R_PU_BOOST_NOT_READY_OD
R54
TNPW040247K0BEED 47000ohm 0.1% 25ppm
R_PD_BUCK_EN
R55
TNPW040247K0BEED 47000ohm 0.1% 25ppm
R_PD_CAP_BOOST_EN
R56
TNPW0402100KBEED 100000ohm 0.1% 25ppm
R_CE_DEFAULT
C33
GRM155R71E104KE14D 100nF 25V X7R
C_BYPASS_U28
C34
GRM155R71E104KE14D 100nF 25V X7R
C_BYPASS_U30
C35
GRM155R71E104KE14D 100nF 25V X7R
C_BYPASS_U31
C36
GRM155R71E104KE14D 100nF 25V X7R
C_BYPASS_U32
C37
GRM155R71E104KE14D 100nF 25V X7R
C_BYPASS_U33
C38
GRM155R71E104KE14D 100nF 25V X7R
C_BYPASS_U34
C39
GRM155R71E104KE14D 100nF 25V X7R
C_BYPASS_U35
C40
GRM155R71E104KE14D 100nF 25V X7R
C_BYPASS_U36
C41
GRM155R71E104KE14D 100nF 25V X7R
C_BYPASS_U37
C42
GRM155R71E104KE14D 100nF 25V X7R
C_BYPASS_U38
C43
GRM155R71E104KE14D 100nF 25V X7R
C_BYPASS_U39
C44
GRM155R71E104KE14D 100nF 25V X7R
C_BYPASS_U17
C45
GRM155R71E104KE14D 100nF 25V X7R
C_BYPASS_U18
AON_2V5
CC_3V3
SYS_3V3
BUCK_EN
CAP_BOOST_EN
GND_USB
SOURCE_SENSE_OD
CAP_OK_OD
CAP_START_N_OD
MAIN_NOT_READY_OD
BOOST_NOT_READY_OD
CHARGE_DISABLE

Sheet: /Actual qualified control/
File: control.kicad_sch

Title: NOVA qualified >=1.5A prototype – control

Size: A1	Date:	**Rev: Q15–PROTOTYPE–1**
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